Pierros Ntelis | Curriculum Vitae

Personal Information	Born on September 18th, 1990 in Athens, Greece Greek citizenship
Contact	NEWMILU, e-mail: ntelis.pierros -at- gmail.com
Research interests	Mathematics, cosmology, theoretical physics, high energy astrophysics, large scale structures - theoretical and observational aspects
Academic	2026-Now School of Physics, Harbin Institute of Technology, Visiting Professor 2025-Now Theoretical Physics at National University of Uzbekistan, Tashkent Senior Researcher (equivalent to Associate Professor) 2022-2026 **International Bilingual School of Provence** Mathematics and Physics Teacher for IB, IGCSE 2018-2020 **Center of Particle Physics of Marseille, Aix University** C.N.E.S. Postdoctoral Research Associate 2014-2017 **Astroparticle and Cosmology Group, Université de Paris** PhD thesis: *Probing Cosmology with the homogeneity scale of the universe through large scale structure surveys*, supervised by Dr. J.C. Hamilton 2013 **National Technical University of Athens** Diploma-Thesis: *Hawking Radiation* supervised by Prof.K.Anagnostopoulos, (6 months) 2012 **National Observatory of Athens** Internship: *Calculation of the age and chemical compositions* of star clusters, supervised by Dr. A. Dapergolas, (2 months)
Education	2014-2017 **Ph.D.**, **Université de Paris, Denis-Diderot** Fundamental Physics, Cosmology, 1st upper class honours 2013-2014 **M.Sc.**, **University Paris 6, Paris 7, Paris 11** Nuclei, Particles, Astroparticles & Cosmology, Grade: $14.9/20$ 2008-2013 **B.A.**, **National Technical University of Athens** School of Applied Mathematical & Physical Science, Grade: $8.5/10$ 2002-2008 **Lyceum of kareas**, **Athens**, Grade: $18.7/20$
Languages	English (fluent), French (intermediate), Greek (native)
Memberships	05/2024- , CosmoVerse network, Cosmic WISPers network 06/2025- , Fundamental challenges in physics theories network 2018-2021, Euclid Consortium, DESI Survey 2014-2017, SDSS-III, BOSS Survey
Awards and Fellowships	2024-2025, 2 CosmoVerse STSM grants E-COST-GRANT-CA21136, (2 week) 2024-2025, Cosmic Whispers YRIG grant E-COST-GRANT-CA21106, (1 week) 2018-2020, C.N.E.S. Post Doctoral Fellowship, (2+1 years) 2014-2017, STEP-UP UnivEarth Sciences, PhD Fellowship 2014 , Paris Denis-Diderot University, Internship Fellowship, (2 months) 2013-2014, Master Award Fellowship, J.S. LATSIS FOUNDATION, (1 year) 2012 , IKY E.S.P.A., Internship Award Fellowship, (2 months)
Publications	ArXiv, ADS, I-HEP, RG, Linkedin, COSMOPIT, ORCID, GS, SP, zM, RL, UT, Z
Interdisciplinary Skills	Maths, efficient communicational skills and learning, data and stat analysis, Mathematical Physics, AI, Python, C/C++, Fortran, Mathematica, LaTeX, HPC, Unix/Linux, CAMB, CLASS, COSMOPIT, mangle, Euclid softs

Teaching & Academic Services	2026, HIT, Harbin, A note on data analysis using Bayes' inference (code), (1h)
	2025, CPT at AMU, Marseille, Advanced Math methods in cosmology, (1h)
	2025, Auckland Univ., New Zealand, Advanced Math methods in cosmology, (1h)
	2025, CPPM, France, Advanced Math methods in cosmology, (1h)
	2025, New U.U., Uzbekinstan, Advanced Math methods in cosmology, (1h)
	2025, SSM, Napoli, STSM: Tensions and Non-Local Analytical Dynamics, (1w)
	2024, ISSA, Malta Univ., STSM: (Non-)Riemannian Cosmic Tensions, (1w)
	2024, Perimeter Institute, Celestial Holography Summerschool participant, (1w)
	2024, Cosmo From Home, Session host in Dynamical System in Cosmology, (1h)
	2023, UCO, Benasque, Host in Modifications of Gravity and Higher-order stats, (5h)
	2023, Cosmo From Home, Session host in Homogeneity, Simulations and LSS, (5h)
	2022, IBS, Mathematics and Physics for High School Students, (6$\times$4 h Seminars, 3y)
	2021 , Euclid , Science talk: Overview of impact of interlopers on PNG, (1h Seminar)
	2020 , Aix-University, From Horndeski to AofEFT, (40' Seminar)
	2020 , Aix-University, M2 intern supervisor on LSS Diagnostics, (25 hours)
	2020 , Aix-University, M1 intern supervisor on BAO/neutrinos, (3 month)
	2020 , Aix-University, M1 intern repporter on Strong Lensing, (1h exam)
	2020 , Aix-University, M1 intern supervisor on BAO, (1 month)
	2020 , DESI, Participation on spectra Visual Inspection (2 months), (2nd Prize in SR)
	2019 , Aix-University, Homogeneity with Multitracers, (2h Seminar)
	2019 , Aix-University, A note on lss, Summer School, (6h Seminar)
	2018 , Renoir, co-organiser of science meetings "Show Plot Club", (weekly, 3y)
	2018 , LSS survey internal reviewer service (DESI, Euclid) (3y)
	2018 , CPPM, LAM, Participation on QC tests of NISP Euclid instrument, (6 months)
	2018 , Aix-University, Scientific Watch, (6h Seminars)
	2018 , SEMFE, NTUA, *Observational Cosmology*, (1.5h Seminar)
	2018 , UiO, What can fractality tell us about the universe?, (1h seminar)
	2018 , CPPM, M1 intership supervisor on BAO, (2 months)
	2018 , Aix-University, A note on lss, Summer School, (3h Seminar)
	2017 , SEMFE, NTUA, *Exploring cosmic homogeneity*, (1.5h Seminar)
	2016-, Referee Report, MNRAS journal
	2015 , Paris-Denis Diderot, Statistical Methods, (4 months)
	2014, APC, Paris 7, co-organiser of cosmology journal club, (weekly, 3y)

Conferences Talks/Posters	2025, The Dark Side of the Universe, FAT dynamics to cosmology (25' Talk)
	2025, Cosmo From Home, Advanced math in cosmology (25' Talk)
	2024, 50 yr of Horndeski at PI, A probabilistic expanding universe under FA, (Poster)
	2024, Cosmo From Home, A probabilistic expanding universe under FA, (25' talk)
	2023, NEB, New avenues of functors of actions theories, (20' Talk)
	2023, Cosmo From Home, New avenues on Functors of Actions, (25' talk)
	2023, NTUA HET, New theories in cosmology and particle physics, (1h Seminar)
	2022, CERN TH DPT Visit, Functors of Actions (5' Blackboard Talk)
	2021, Euclid Science Talk, Overview of impact of interlopers on PNG (1h Talk)
	2020, Atelier D.E., PNG with interlopers in Euclid, (20' Talk)
	2020, Atelier D.E., Perturbation Theory of LSS, code: CLASS-PT, (20' Talk)
	2020, SEMFE, NTUA, LSS: Theory and Observations, (40' Seminar)
	2019, 30th SYMPOSIUM, Fractality from Multiple Tracers, (15' Talk)
	2019, Euclid IAS Paris, Systematics on PNG, (20' Talk)
	2019, Euclid Consortium Helnsinki, Primordial Non-Gaussianity, (15' Talk)
	2019, eBOSS/DESI France, Cosmology with Cosmic Homogeneity (15' Talk)
	2018, Euclid GC-WL, Ameliorating cosmology with homogeneity, (15' Talk)
	2018, Colloquium of D.E., Cosmic Homogeneity as a standard ruler, (20' Talk)
	2018, R. Moriond, *Euclid:Homogeneity in search of the Dark Sector*, (Poster)
	2017, P.O.N.T. in Cosmology, *Is the universe homogeneous?*, (20' Talk)
	2016, R. Moriond, *The homogeneity scale of the universe*, (15' Talk)
	2015, UnivEarthS School, Santorini, *LSS and homogeneity*, (Poster)
	2015, Advance School of Cosmology at I.C.T.P., Trieste
	2015, Rencontres de Jeunes Physicienne, *Homogeneity* (Video), (15' Talk)
	2015, IDPASC school, at LPNHE, Paris
	2014, Elbereth Conference, IAP, *The homogeneity of the universe* (15' Talk)
	2014, UnivEarthS School, *The homogeneity of the universe*, (3' Talk)
Certificates	Qualif. of « Maitre de conférences » (sec. 26 "Applied Math") obtained 2026
	Qualif. of « Maitre de conférences » (sec. 29 "Elementaries") obtained 2019
	IB Mathematics: Applications and Interpretations
Outreach Talks	2024-Today, Utube, Introduction cosmology for everyone (1h talk)
	2023('24), IBS of Provence, Neuroscience and cosmology for high school students (3h)
	2023 , IBS of Provence, Math and physics from CPT for high school students (3h)
	2022 , IBS of Provence, Mathematics and physics for high school students (3h)
Other Interests	creative writing, piano, guitar, running, water sports, skiing

List of publication and products

Dr. Pierros Ntelis

April 20, 2026

Via, an h-index: 14 and total citations: 3327 (Google Scholar 2025), my most important work is represented in the following papers:

Refereed

[1] P. Yovkochev and ***Ntelis, P.*** , 2026. Evolution and Mass Decay of a Gaussian Pulse in Time-Fractional Diffusion: An L1-Implicit Scheme Analysis. *Turanian Journal*, vol. 2, no. 1, pp. 020103-1–020103-7. URL, Contribution: Supervision, and co-design of analysis. (Citations: 0)

[2] ***Ntelis, P.*** 2025. Advancing Tensor Theories. Symmetry: Mathematics: Advances in Topology and Algebraic Geometry. URL (Citations: 2) ⋆

[3] ***Ntelis, P.*** 2025. Advanced manifold-metric pairs. Mathematics. URL (Citations: 1) ⋆

[4] ***Ntelis, P.*** & J. L. Said, 2025. Analytical *poly*ΛCDM dynamics. PDU.URL (Citations: 1) ⋆

[5] E.D. Valentino, J. L. Said, ..., ***Ntelis, P.*** ..., et al. 2025. The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics. Contribution: Writing on extended theories of gravity, scientific review, and suggestions. PDU URL (Citations: 97)

[6] Ntelis, P. 2022. A (D_τ, D_x)-manifold with N-correlators of N_t-objects. IJGMMP, arXiv e-prints arXiv:2209.07472 URL, (Citations: 0)

[7] ***Ntelis, P.*** & J. L. Said 2025. Exploring ϕCDM model dynamics. EPJC, arXiv:2502.03486 URL, (Citations: 4)

[8] ***Ntelis, P.*** & J. L. Said 2025. Simple $\phi\Lambda$CDM dynamics. IJGMMP, URL, (Citations: 3)

[9] S.Casas, ..., ***Ntelis, P.*** , and 128 colleagues 2025. Euclid: Constraints on f(R) cosmologies from the spectroscopic and photometric primary probes. A&A, URL (Citations: 27) Contribution: Scientific review, software development.

[10] Euclid Collaboration, and 1137 colleagues 2024. Euclid. I. Overview of the Euclid mission. A&A, arXiv:2405.13491 URL, (Citations: 277) Contribution: Scientific review

[11] ***Ntelis, P.*** & Morris, A. 2023. Functors of actions. Foundations of Physics, arXiv:2010.06707 URL, (Citations: 18) ⋆

[12] Göksel Karaçaylı, N., ... ***Ntelis, P.*** , and 10 colleagues 2022. Optimal 1D Lyα Forest Power Spectrum Estimation – II. KODIAQ, SQUAD & XQ-100. MNRAS URL, (Citations: 54), Contribution: Scientific review and contribution on discussions for further research and software development.

[13] Hamaus, N., ... ***Ntelis, P.*** , and 128 colleagues 2022. Euclid: Forecasts from redshift-space distortions and the Alcock-Paczynski test with cosmic voids. A&A arXiv e-prints URL, (Citations: 82), Contribution: Scientific review and contribution on discussions for further research and software development.

[14] Ilić, S., ... ***Ntelis, P.*** , and 207 colleagues 2021. *Euclid* preparation: XV. Forecasting cosmological constraints for the *Euclid* and CMB joint analysis. A&A arXiv e-prints URL, (Citations: 74), Contribution: Scientific review and contribution on discussions for further research. ⋆

[15] Alam, S., ... ***Ntelis, P.*** , and 33 colleagues 2020. Towards testing the theory of gravity with DESI: summary statistics, model predictions and future simulation requirements. JCAP URL, (Citations: 57), Contribution: Scientific review and contribution on discussions for further research and software development. ⋆

[16] Tutusaus, I., ... ***Ntelis, P.*** , and 89 colleagues 2020. Euclid: The importance of galaxy clustering and weak lensing cross-correlations within the photometric Euclid survey. Astronomy and Astrophysics 643, URL, (Citations: 78), Contribution: Scientific review and contribution on discussions for further research.

[17] ***Ntelis, P.*** , A. Ealet, S. Escoffier, J.-C. Hamilton, A. J. Hawken, J.-M. Le Goff, J. Rich, and A. Tilquin 2018. The scale of cosmic homogeneity as a standard ruler. Journal of Cosmology and Astroparticle Physics 2018, URL, (Citations: 23)

[18] Blanton, M. R., ... ***Ntelis, P.*** , and 361 colleagues 2017. Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe. The Astronomical Journal 154, URL, (Citations: 1984), Contribution: Scientific review and contribution on discussions for further research. ⋆

[19] ***Ntelis, P.*** , and 22 colleagues 2017. Exploring cosmic homogeneity with the BOSS DR12 galaxy sample. Journal of Cosmology and Astroparticle Physics 2017, URL, (Citations: 101) ⋆

[20] Laurent, P., ... ***Ntelis, P.*** , and 17 colleagues 2016. A 14 h^{-3} Gpc^3 study of cosmic homogeneity using BOSS DR12 quasar sample. Journal of Cosmology and Astroparticle Physics 2016, URL, (Citations: 89), Contribution: Scientific review and contribution on discussions for further research and software development.

Under review

[1] ***Ntelis, P.*** , et al. 2019. Cosmological constraints from cosmic homogeneity. (Citations: 4) URL

[2] ***Ntelis, P.*** 2025. A probabilistic and expanding Universe. URL, (Citations: 2)

[3] ***Ntelis, P.*** 2023. Advancing category and tensor theories. URL, (Citations: 2)

[4] ***Ntelis, P.*** 2025. CosmoFATs. URL, (Citations: 2)

[5] ***Ntelis, P.*** 2025. CosmoFATs Dynamics. URL (Citations: 0)

[6] ***Ntelis, P.*** 2025. A cosmic Schiff's conjecture test. URL, (Citations: 0)

[7] ***Ntelis, P.*** 2025. A universal foundational theory. URL, (Citations: 0)

[8] ***Ntelis, P.*** 2026. · A dot. URL, DOI, (Citations: 0)

Outreach

2022-t_{now}, Universions Youtube Channel
2025 , Utube, Comet 3i ATLAS: Interstellar Halloween Secrets! (6')
2025 , Utube, Are there Aliens ? Drake's Equation ! (7')
2025 , Utube, Quantum Tunnelling in Your Palm! (1')
2025 , Utube, Advanced Manifold-Metric Pairs (10')
2025 , Utube, A sunny watery magic (33")
2025 , Utube, Advanced Euler's identity (3')
2025 , Utube, Analytical $poly\Lambda$CDM dynamics 101 (5' talk)
2025 , Utube, Advancing Tensor Theories 101 (10' talk)
2025 , Utube, Quintesssence 101 (5' talk)
2024 , Utube, Introduction cosmology for everyone (1h talk)
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	2019, 30th SYMPOSIUM, Fractality from Multiple Tracers, (15' Talk)
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	2019, Euclid Consortium Helnsinki, Primordial Non-Gaussianity, (15' Talk)
	2019, eBOSS/DESI France, Cosmology with Cosmic Homogeneity (15' Talk)
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	2018, Euclid GC-WL, Ameliorating cosmology with homogeneity, (15' Talk)
	2018, Colloquium of D.E., Cosmic Homogeneity as a standard ruler, (20' Talk)
	2018, R. Moriond, *Euclid:Homogeneity in search of the Dark Sector*, (Poster) HIC
	2018, UiO, What can fractality tell us about the universe?, (1h seminar)
	2017, P.O.N.T. in Cosmology, *Is the universe homogeneous?*, (20' Talk)
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	2014, Elbereth Conference, IAP, *The homogeneity of the universe* (15' Talk)
	2014, UnivEarthS School, *The homogeneity of the universe*, (3' Talk)

Software

[1] P.Ntelis (2017-today): Github contributions
[2] P.Ntelis (2017-today): COSMOPIT
[3] P.Ntelis (2018-2021): Gitlab private Euclid softwares
[4] P.Ntelis (2018): MCMC-posterior-example
[5] P.Ntelis (2018): CoHo
[6] P.Ntelis (2018): ANALYSE-SN-magnitude-redshift-curve
[7] P.Ntelis (2018): Extract-Cosmology-with-the-BAO-peak-position
[8] P.Ntelis (2020): AofEFT
[9] P.Ntelis (2023): Tutorial on particle/galaxy simulations and coordinate transformation
[10] P.Ntelis (2025a): Standard ϕCDM dynamics
[11] P.Ntelis (2025b): Simple quintessence dynamics
[12] P.Ntelis (2025c): Analytical $poly\Lambda$CDM dynamics